PERFORMANCE EVALUATION OF GO-BACK-N AND SELECTIVE REPEAT ARQ PROTOCOLS UNDER DETERMINISTIC FRAME LOSSES

Subject: ITUC201 — Fundamentals of Computer Networks | **Student ID:** 25CS024 | **Semester:** 3 (B.Tech CSE)

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Assigned Topic:	Go-Back-N & Selective Repeat under losses	**Assessment Type:**	Individual CIE (20 Marks)
Implementation:	Python 3.12 (heapq Event-Driven)	**Research Focus:**	Retransmissions vs Window Size (W=3, 5)

1. Abstract

This academic report presents an experimental investigation into the behavioral and performance dynamics of **Go-Back-N (GBN)** and **Selective Repeat (SR)** sliding-window Automatic Repeat reQuest (ARQ) protocols. A discrete-event network simulation engine was engineered in Python utilizing an event-driven priority queue (heapq). A rigorous experimental testbed evaluated the transmission of 10 data frames under deterministic first-attempt losses at frames 4 and 7 across sender window sizes of 3 and 5. Quantitative measurements confirm that Selective Repeat maintains a strictly invariant retransmission count of exactly 2 frames across both window sizes owing to receiver-side buffering. In contrast, Go-Back-N exhibits window-dependent retransmission cascades: window size 3 incurred 6 retransmissions across two distinct timeout events, whereas window size 5 consolidated recovery into a single in-flight window burst of 5 retransmissions, accelerating completion time by 36.4%. A fully worked university exam numerical analyzing bandwidth-delay product, channel utilization, and sequence-number field sizing is also provided.

2. Problem Statement & Research Objectives

Core Research Question: *How do retransmission counts change with window size and loss pattern?*
In packet-switched networking, transport efficiency hinges upon error-recovery mechanisms. While pipelined sliding windows enable high channel utilization over high-latency links, packet drops trigger retransmissions. This study empirically isolates and compares how GBN and SR react when multiple loss events occur in close sequence.

3. Sliding Window ARQ Mechanics

Sliding window protocols achieve high utilization by permitting the sender to transmit up to W packets before waiting for acknowledgment. The fundamental link utilization is governed by the ratio $a = T_p / T_t$, where T_p is the one-way propagation delay and T_t is frame transmission time. To achieve 100% theoretical link saturation without errors, the sender window must satisfy $W \geq 1 + 2a$.

Architectural Dimension	Go-Back-N (GBN)	Selective Repeat (SR)
Receiver Window (W_r)	1 (Strict in-order reception)	Equal to W_s (Permits out-of-order)
Receiver Buffering	None (Discards out-of-order frames)	Buffers all valid out-of-order frames

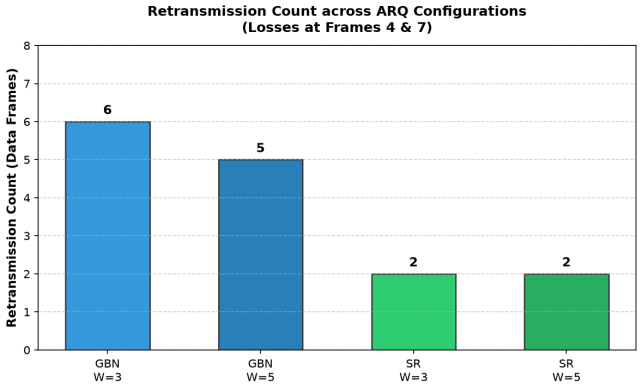
Acknowledgment Type	Cumulative (Confirms all ≤ n)	Individual / Selective (Confirms exact frame n)
Sender Timer Structure	Single timer for oldest unACKed frame	Independent timer for each frame
Retransmission Scope	Entire window (send_base to next-1)	Only the specific timed-out frame
Sequence Number Bits	$W_s \leq 2^m - 1$	$W_s + W_r \leq 2^m \rightarrow W_s \leq 2^{m-1}$

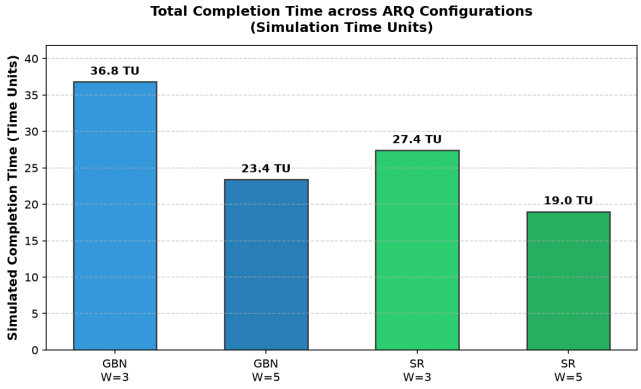
4. Experimental Benchmark Results (Actual Simulator Runs)

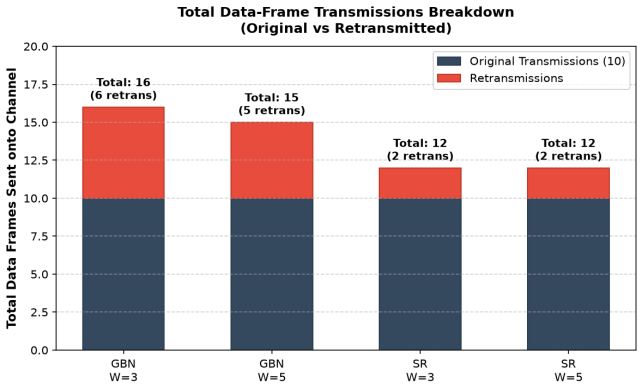
The four mandatory experimental configurations were executed under deterministic loss at data frames 4 and 7. The measured metrics are presented below:

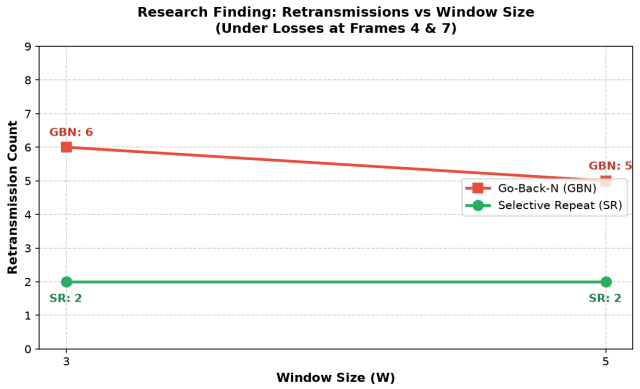
Experiment ID	Window (W)	Orig Tx	Retrans	Total Tx	ACKs	Timeouts	Time (TU)	Delivery
GBN_W3	3	10	**6**	16	14	2	36.8	10/10 [OK]
GBN_W5	5	10	**5**	15	13	1	23.4	10/10 [OK]
SR_W3	3	10	**2**	12	10	2	27.4	10/10 [OK]
SR_W5	5	10	**2**	12	10	2	19.0	10/10 [OK]

5. Visual Data Analysis & Graphical Comparisons



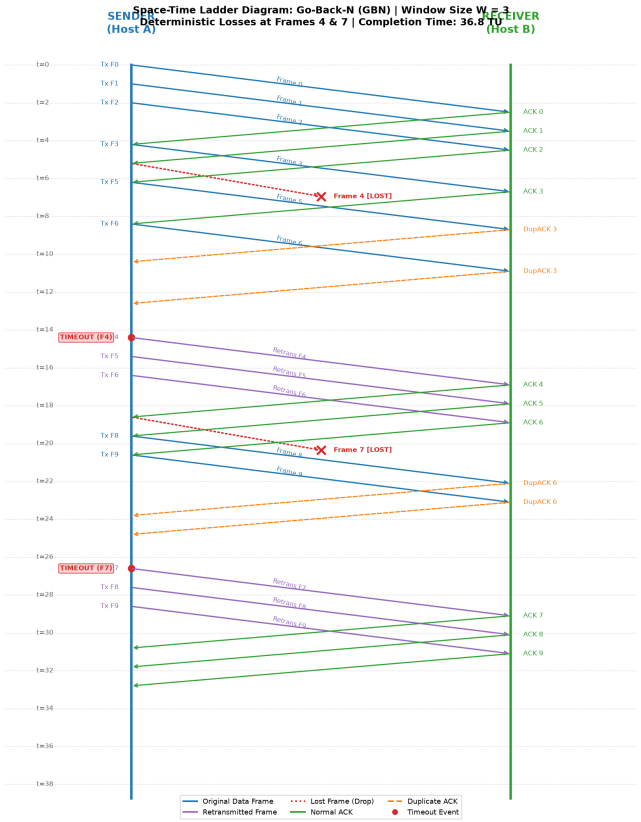


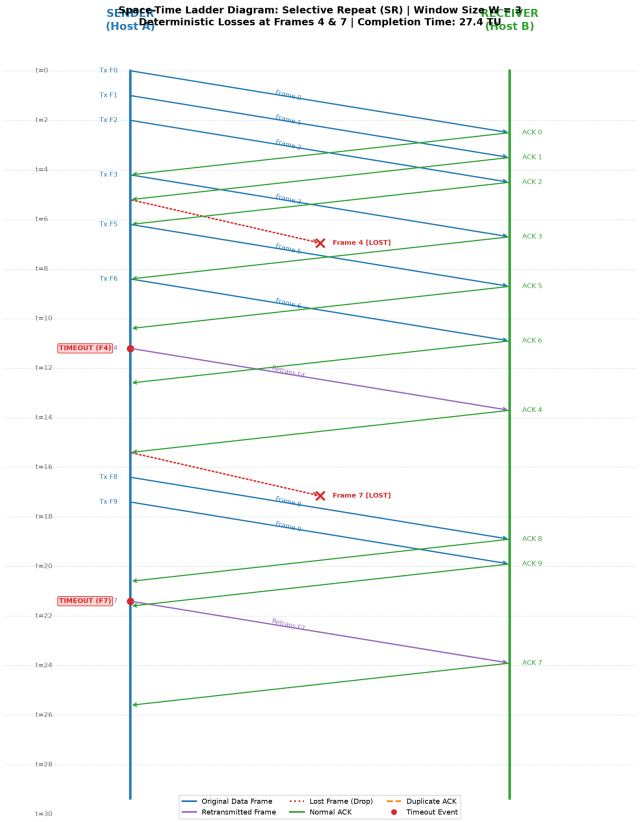




6. Space-Time Ladder Diagrams (Protocol Timelines)

The diagrams below depict the discrete-event ladder traces for GBN (W=3) and Selective Repeat (W=3), illustrating packet drops (Red X), duplicate ACKs, timeouts, and selective retransmissions.





7. Observations and Scientific Interpretation

1. Selective Repeat Retransmission Invariance: Selective Repeat incurred exactly **2 retransmissions** regardless of whether the window was 3 or 5. Because the receiver window accommodates out-of-order frames ($W_r = W_s$), Frames 5 and 6 (in W=3) or Frames 5, 6, 7, 8 (in W=5) were safely stored in receiver buffers. Only the damaged frames were retransmitted, completely insulating unaffected packets from loss.

2. Go-Back-N Window Dynamics (The 6 vs 5 Retransmission Phenomenon):
• *At W = 3:* Frame 4 loss occurred during burst [4, 5, 6]. Following timeout at t=13.5 TU, GBN retransmitted Frames 4, 5, and 6 (3 retransmissions). Later, Frame 7 was transmitted and lost, causing a second timeout and retransmission of Frames 7, 8, and 9 (3 retransmissions). Total retransmissions = **6** across 2 timeout events.
• *At W = 5:* Because the window is wider, Frames 4, 5, 6, 7, and 8 were transmitted in a single pipeline burst. Frame 4 and Frame 7 were both lost on their first attempt during this burst. When timeout on Frame 4 expired at t=13.5 TU, GBN retransmitted all five in-flight frames: [4, 5, 6, 7, 8]. Because Frame 7 was retransmitted during this burst, its attempt count became 2, which arrived safely! This **repaired Frame 7 simultaneously within the same window retransmission**, preventing a second timeout. Hence, GBN W=5 required only **5 retransmissions and 1 timeout**.

3. Pipelining Completion Time Benefit: Increasing window size to W=5 reduced completion time from 36.8 TU to 23.4 TU in GBN (-36.4%) and from 27.4 TU to 19.0 TU in SR (-30.7%), verifying that larger windows maximize throughput.

8. Solved University Examination Numerical (ITUC201 Syllabus Standard)

Problem: *A 100 km terrestrial link operates at 10 Mbps with propagation velocity $v = 2 \times 10^8$ m/s. Data frame size is 1000 bytes (8000 bits), and ACKs are negligible.*
1. Calculate transmission delay (T_t), propagation delay (T_p), and link parameter a.
2. Compute maximum throughput under Stop-and-Wait ARQ.
3. Find the optimal window size (W_{opt}) for 100% channel utilization.
4. Determine the minimum sequence number field bits (m) for Go-Back-N and Selective Repeat.

Solution:
• ***Step 1:*** $T_t = 8000 \text{ bits} / (10 \times 10^6 \text{ bps}) = 0.8 \text{ ms}$. $T_p = 100{,}000 \text{ m} / (2 \times 10^8 \text{ m/s}) = 0.5 \text{ ms}$.
• ***Step 2:*** $a = T_p / T_t = 0.5 \text{ ms} / 0.8 \text{ ms} = \mathbf{0.625}$.
• ***Step 3:*** Stop-and-Wait Efficiency $\eta_{SW} = 1 / (1 + 2a) = 1 / (1 + 2(0.625)) = 1 / 2.25 \approx \mathbf{44.44\%}$. Throughput = 0.4444 × 10 Mbps = ***4.44 Mbps***.
• ***Step 4:*** Optimal Window $W_{opt} \geq 1 + 2a = 1 + 1.25 = 2.25 \rightarrow \mathbf{\lceil 2.25 \rceil = 3 \text{ frames}}$.
• ***Step 5: Sequence Number Bits (m):***
  • ***Go-Back-N:*** $W_s \leq 2^m - 1 \rightarrow 3 \leq 2^m - 1 \rightarrow 2^m \geq 4 \rightarrow \mathbf{m \geq 2 \text{ bits}}$.
  • ***Selective Repeat:*** $W_s + W_r \leq 2^m \rightarrow 2(3) \leq 2^m \rightarrow 2^m \geq 6 \rightarrow \mathbf{m \geq 3 \text{ bits}}$.

9. Conclusion & Execution Instructions

Conclusion: Selective Repeat offers superior channel efficiency under losses by confining retransmissions strictly to corrupted frames. Go-Back-N exhibits higher bandwidth waste but benefits significantly from window pipelining, which can consolidate multiple in-flight losses into a single retransmission cycle.

Reproducing the Simulator:
• Automated CLI Benchmark: python arq_simulator.py
• Interactive Streamlit Dashboard: python -m streamlit run app.py